



PART 1 — Linux File & Folder Commands

काम	Command	क्या करेगा	
Current location	<code>pwd</code>	Current directory का पूरा path	
Files/folders देखना	<code>ls</code>	Current directory की contents	
Detailed list	<code>ls -l</code>	Permissions, owner, size, date आदि	
Hidden files	<code>ls -la</code>	Hidden files सहित	
Folder create	<code>mkdir backend</code>	<code>backend</code> folder बनाएगा	
Multiple folders	<code>mkdir backend frontend config models</code>	कई folders एक साथ	
Multiple folders {} से	<code>mkdir {backend,frontend,config,models}</code>	Brace expansion से folders	
Nested folders	<code>mkdir -p</code>	Parent + अंदर के folders	



Linux command

Nested folders	<code>mkdir -p backend/{config,controllers,model s,routes}</code>	Parent + अंदर के folders
Empty file	<code>touch app.js</code>	Empty file बनाएगा
Multiple files	<code>touch app.js server.js db.js</code>	कई files बनाएगा
Multiple files {} से	<code>touch {app,server,db}.js</code>	<code>app.js</code> , <code>server.js</code> , <code>db.js</code>
File देखना	<code>cat app.js</code>	File का पूरा content
File की शुरुआत	<code>head app.js</code>	शुरुआत की lines
File का अंत	<code>tail app.js</code>	आखिरी lines
Live log	<code>tail -f app.log</code>	नई log lines live दिखाता रहेगा
File copy	<code>cp app.js backup.js</code>	File copy
Folder copy	<code>cp -r backend backup</code>	पूरा folder copy

Linux command

Folder copy	<code>cp -r backend backup</code>	पूरा folder copy
File move	<code>mv app.js backend/</code>	File backend में
Folder move	<code>mv backend frontend/</code>	backend को frontend में
Rename file	<code>mv app.js main.js</code>	File rename
Rename folder	<code>mv backend server</code>	Folder rename
File delete	<code>rm app.js</code>	File delete
Multiple files delete	<code>rm app.js server.js db.js</code>	कई files delete
Folder delete	<code>rmdir backend</code>	Empty folder delete
पूरा folder delete	<code>rm -r backend</code>	Folder + content delete
Force delete	<code>rm -rf backend</code>	बिना confirmation recursively delete

PART 2 — Multiple Folders Create

Without loop

 Bash

```
mkdir backend frontend config models routes controllers middleware
```

expansion से

 Bash

```
mkdir {backend,frontend,config,models,routes,controllers,middleware}
```

Nested folders

 Bash

```
mkdir -p backend/{config,controllers,middleware,models,routes,utils}
```



PART 3 — Multiple Folders Delete

Empty folders

</> Bash



```
rm -d backend frontend config
```

Content सहित

</> Bash



```
rm -r backend frontend config
```

Force + recursive

</> Bash



```
rm -rf backend frontend config
```

⚠️ `rm -rf` बहुत dangerous command है। ↓ path देने पर important data delete हो सकता है।

Linux command

```
Bash  
  
for i in backend frontend config models  
do  
    mkdir "$i"  
done
```

एक line में:

```
</> Bash  
  
for i in backend frontend config models; do mkdir "$i"; done
```

Delete loop

```
</> Bash  
  
for i in backend frontend config models  
do  
    rm -r "$i"  
done
```



PART 5 — Multiple Files

बिना loop

</> Bash

```
touch app.js server.js db.js auth.js
```

Loop

</> Bash

```
for i in app.js server.js db.js auth.js
do
    touch "$i"
done
```

एक line:

</> Bash



```
for i in app.js server.js db.js auth.js; do touch "$i"; done
```

Delete बिना loop

</> Bash

```
rm app.js server.js db.js auth.js
```

Delete loop से

</> Bash

```
for i in app.js server.js db.js auth.js
do
    rm "$i"
done
```



PART 6 — Different Extensions

Without loop

</> Bash

```
touch app.js style.css index.html data.json README.md
```

Result:

```
app.js
style.css
index.html
data.json
README.md
```

Brace expansion

</> Bash

```
touch app.js style.css index.html data.json README.md
```



PART 14 — Linux में सबसे जरूरी Basic Commands

Command	काम	Example
<code>pwd</code>	Current location	<code>pwd</code>
<code>ls</code>	Files/folders	<code>ls</code>
<code>ls -la</code>	Hidden + detailed	<code>ls -la</code>
<code>cd</code>	Directory change	<code>cd backend</code>
<code>cd ..</code>	Parent directory	<code>cd ..</code>
<code>cd ~</code>	Home directory	<code>cd ~</code>
<code>mkdir</code>	Folder create	<code>mkdir project</code>
<code>mkdir -p</code>	Nested folders	<code>mkdir -p a/b/c</code>
<code>touch</code>	File create	<code>touch app.js</code>

Linux command

<code>touch</code>	File create	<code>touch app.js</code>
<code>cp</code>	Copy	<code>cp app.js backup.js</code>
<code>cp -r</code>	Folder copy	<code>cp -r backend backup</code>
<code>mv</code>	Move/Rename	<code>mv app.js main.js</code>
<code>rm</code>	File delete	<code>rm app.js</code>
<code>rm -r</code>	Folder recursively delete	<code>rm -r backend</code>
<code>cat</code>	File content	<code>cat app.js</code>
<code>less</code>	File पढ़ना	<code>less app.log</code>
<code>head</code>	शुरुआत की lines	<code>head app.log</code>
<code>tail</code>	आखिरी lines	<code>tail app.log</code>
<code>tail -f</code>	Live logs	<code>tail -f app.log</code>



<code>tail -f</code>	Live logs	<code>tail -f app.log</code>
<code>clear</code>	Terminal clear	<code>clear</code>
<code>history</code>	Previous commands	<code>history</code>
<code>man</code>	Manual/help	<code>man ls</code>



PART 15 — Search Commands

Command	काम	Example
<code>find</code>	Files/folders खोजना	<code>find . -name "app.js"</code>
<code>grep</code>	Text search	<code>grep "hello" app.js</code>
<code>grep -r</code>	Recursively text search	<code>grep -r "mongoose" .</code>
<code>which</code>	Command का path	<code>which node</code>



Linux command

<code>grep</code>	Text search	<code>grep "hello" app.js</code>
<code>grep -r</code>	Recursively text search	<code>grep -r "mongoose" .</code>
<code>which</code>	Command का path	<code>which node</code>
<code>whereis</code>	Binary/source/man locations	<code>whereis node</code>
<code>locate</code>	File search database से	<code>locate app.js</code>

बहुत useful:

⌕ Bash

```
grep -r "PORT" .
```

Current directory के अंदर recursively `PORT` search करेगा।



PART 16 — File Information

Command	काम
<code>file app.js</code>	File type बताता है
<code>wc app.js</code>	Lines/words/bytes
<code>wc -l app.js</code>	सिर्फ lines
<code>du -sh backend</code>	Folder का approximate disk usage
<code>df -h</code>	Disk filesystem usage
<code>ls -lh</code>	Human-readable file sizes
<code>stat app.js</code>	Detailed file metadata



PART 18 — User Commands

Command	काम	Example
<code>whoami</code>	Current user	<code>whoami</code>
<code>id</code>	User/group IDs	<code>id</code>
<code>users</code>	Logged-in users	<code>users</code>
<code>who</code>	Logged-in sessions	<code>who</code>
<code>w</code>	Users + activity	<code>w</code>
<code>passwd</code>	Password change	<code>passwd</code>
<code>sudo</code>	Root privileges	<code>sudo apt update</code>
<code>su</code>	User switch	<code>su username</code>



PART 19 — Process Commands

Backend developer के लिए बहुत important!

Command	काम	Example
<code>ps</code>	Processes	<code>ps</code>
<code>ps aux</code>	सभी processes की detailed list	<code>ps aux</code>
<code>top</code>	Live process monitor	<code>top</code>
<code>htop</code>	Better interactive process monitor	<code>htop</code>
<code>kill</code>	PID से process terminate	<code>kill 1234</code>
<code>kill -9</code>	Force kill	<code>kill -9 1234</code>
<code>pkill</code>	Name से process kill	<code>pkill node</code>
<code>pgrep</code>	Name से PID खोजना	<code>pgrep node</code>





PART 20 — Linux Networking

Command	काम	Example
<code>ip addr</code>	Network interfaces/IP	<code>ip addr</code>
<code>ip route</code>	Routing table	<code>ip route</code>
<code>ping</code>	Connectivity	<code>ping google.com</code>
<code>ss</code>	Sockets/ports	<code>ss -tulpn</code>
<code>curl</code>	HTTP request	<code>curl https://example.com</code>
<code>wget</code>	Download	<code>wget URL</code>
<code>dig</code>	DNS lookup	<code>dig google.com</code>
<code>nslookup</code>	DNS lookup	<code>nslookup google.com</code>
<code>traceroute</code>	Route/hops	<code>traceroute google.com</code>





PART 1 — 8 Images में मिली Linux Commands

[Share](#)

#	Image में Command	क्या काम करती है	Command का Breakdown	Example / Result	
1	<code>yum update</code>	Linux packages को update करता है	<code>yum</code> = package manager, <code>update</code> = packages update	<code>sudo yum update</code>	
2	<code>yum install git</code>	Git package install करता है	<code>install</code> = package installation	<code>sudo yum install git</code>	
3	<code>vi app</code>	<code>app</code> नाम की file को <code>vi</code> editor में खोलता/बनाता है	<code>vi</code> = terminal text editor	<code>vi app</code>	
4	<code>cat app</code>	File का content terminal में दिखाता है	<code>cat</code> = concatenate/display	<code>cat app</code>	
5	<code>history</code>	पहले चलाए गए <code>commands</code> दिखाता है	shell command history	<code>history</code>	
6	<code>sudo apt update</code>	Package repositories की latest information fetch करता है	<code>sudo</code> = elevated privileges, <code>apt</code> = package manager	<code>sudo apt update</code>	
7	<code>sudo apt-get update -y</code>	Package information update	<code>apt-get</code> = package manager, <code>-y</code> = automatic yes	<code>sudo apt-get update -y</code>	
8	<code>sudo apt install fontconfig openjdk-21-jre -y</code>	Java runtime + fontconfig install	<code>install</code> + packages + <code>-y</code>	Java install	
9	<code>java -version</code>	Installed Java version check	<code>java</code> executable + <code>-version</code>	Java version output	
10	<code>sudo mkdir -p /etc/apt/keyrings</code>	Directory बनाता है	<code>mkdir</code> = make directory, <code>-p</code> = parent directories भी	<code>/etc/apt/keyrings</code>	

Linux command

Share

11	<code>sudo wget -O ... URL</code>	Internet से file download करता है	<code>wget</code> = downloader, <code>-O</code> = output filename/path	Jenkins key download
12	<code>echo "..."</code>	Text terminal में print करता है	<code>echo</code> = output text	<code>echo "hello"</code>
13	<code>sudo tee /etc/apt/sources.list.d/jenkins.list</code>	Input को file में लिखता है	<code>tee</code> = stdin → file/output	Repository file
14	<code>> /dev/null</code>	Output को discard करता है	<code>></code> = redirection, <code>/dev/null</code> = discard location	Output दिखाई नहीं देगा
15	<code>sudo apt update -y</code>	Package lists update	<code>apt update</code> + automatic yes option	Repository refresh
16	<code>sudo apt install jenkins -y</code>	Jenkins package install	<code>apt install</code>	Jenkins install
17	<code>systemctl status jenkins</code>	Jenkins service का status	<code>systemctl</code> = service manager	active (running)
18	<code>cd ..</code>	Parent directory में जाता है	<code>cd</code> = change directory, <code>..</code> = parent	<code>/home/user/project</code> → <code>/home/user</code>
19	<code>sudo cat /var/lib/jenkins/secrets/initialAdminPassword</code>	Jenkins की initial admin password file पढ़ता है	<code>sudo</code> + <code>cat</code>	Password terminal में
20	<code>clear</code>	Terminal screen साफ करता है	<code>clear</code>	Screen clean
21	<code>apt update</code>	Package repository information update	<code>apt</code> package manager	<code>sudo apt update</code>

Linux command

21	<code>apt update</code>	Package repository information update	<code>apt</code> package manager	<code>sudo apt update</code>
22	<code>apt install</code>	Packages install	package installation	<code>sudo apt install nginx</code>
23	<code>cat</code>	File content पढ़ना	file → terminal	<code>cat file.txt</code>
24	<code>vi</code>	Terminal editor	edit/create file	<code>vi app.js</code>
25	<code>sudo</code>	Command को elevated privileges से चलाता है	<code>sudo</code> = superuser do	<code>sudo apt update</code>

1. `yum` — Package Manager

Images में:

`</>` **Bash**

```
sudo yum update
sudo yum install git
```



`yum` क्या है?

`yum` Linux में software packages install/update/manage करने वाला package manager है।

```
yum
↓
Package Manager
↓
Install / Update / Remove software
```



Linux command

Command	काम
<code>yum update</code>	Packages update
<code>yum install git</code>	Git install
<code>yum remove git</code>	Git remove
<code>yum search git</code>	Package search
<code>yum list installed</code>	Installed packages
<code>yum info git</code>	Package information

Ubuntu/Debian systems में `apt` package management के लिए बहुत important है।

Command	काम
<code>sudo apt update</code>	Repository package lists refresh
<code>sudo apt upgrade</code>	Installed packages upgrade
<code>sudo apt install nginx</code>	Package install
<code>sudo apt remove nginx</code>	Package remove
<code>sudo apt purge nginx</code>	Package + config files remove करने की कोशिश
<code>apt search nginx</code>	Package search
<code>apt show nginx</code>	Package information
<code>apt list --installed</code>	Installed packages
<code>sudo apt autoremove</code>	Unneeded dependencies remove

vi के basic modes

Key	काम
i	Insert mode
Esc	Normal mode
:w	Save
:q	Quit
:wq	Save + quit
:q!	बिना save quit

Basn

```
systemctl status jenkins
```

Linux services manage करने के लिए बहुत important command I

Command	काम
systemctl status jenkins	Status
sudo systemctl start jenkins	Start
sudo systemctl stop jenkins	Stop
sudo systemctl restart jenkins	Restart
sudo systemctl enable jenkins	Boot पर auto-start
sudo systemctl disable jenkins	Auto-start disable



PART 3 — File पढ़ना/Edit करना

Command	काम	Example
<code>cat</code>	पूरा content	<code>cat app.js</code>
<code>less</code>	Page-wise पढ़ना	<code>less app.log</code>
<code>more</code>	Page-wise पढ़ना	<code>more app.log</code>
<code>head</code>	Beginning की lines	<code>head app.log</code>
<code>tail</code>	Last lines	<code>tail app.log</code>
<code>tail -f</code>	Live log	<code>tail -f app.log</code>
<code>nano</code>	Simple editor	<code>nano app.js</code>
<code>vi</code>	Terminal editor	<code>vi app.js</code>
<code>vim</code>	Advanced vi	<code>vim app.js</code>



PART 4 — Search

Command	काम	Example
<code>find</code>	Files/folders search	<code>find . -name "app.js"</code>
<code>grep</code>	Text search	<code>grep "PORT" app.js</code>
<code>grep -r</code>	Recursive text search	<code>grep -r "mongoose" .</code>
<code>which</code>	Command का executable path	<code>which node</code>
<code>whereis</code>	Binary/info locations	<code>whereis node</code>



PART 5 — Process Commands

MERN/Node के लिए बहुत important:

Command	काम	Example
<code>ps</code>	Processes	<code>ps</code>
<code>ps aux</code>	सभी processes	<code>ps aux</code>
<code>top</code>	Live process monitoring	<code>top</code>
<code>kill</code>	PID से process terminate	<code>kill 1234</code>
<code>kill -9</code>	Force terminate	<code>kill -9 1234</code>
<code>pgrep</code>	Process name से PID	<code>pgrep node</code>
<code>pkill</code>	Name से processes terminate	<code>pkill node</code>



PART 6 — Network Commands

Command	काम	Example
<code>ip addr</code>	IP/interfaces	<code>ip addr</code>
<code>ip route</code>	Routing table	<code>ip route</code>
<code>ping</code>	Connectivity	<code>ping google.com</code>
<code>ss</code>	Ports/sockets	<code>ss -tulpn</code>
<code>curl</code>	HTTP request	<code>curl http://localhost:5000</code>
<code>wget</code>	Download	<code>wget URL</code>
<code>dig</code>	DNS lookup	<code>dig google.com</code>
<code>hostname</code>	Hostname	<code>hostname</code>



PART 7 — Permissions

Command	काम	Example
<code>ls -l</code>	Permissions देखना	<code>ls -l</code>
<code>chmod</code>	Permissions change	<code>chmod +x app.sh</code>
<code>chown</code>	Owner change	<code>sudo chown user file</code>
<code>chgrp</code>	Group change	<code>sudo chgrp developers file</code>
<code>sudo</code>	Elevated privileges	<code>sudo apt update</code>
<code>id</code>	User/group IDs	<code>id</code>
<code>whoami</code>	Current user	<code>whoami</code>

PART 8 — Disk Commands

Command	काम	Example
<code>df -h</code>	Disk space	<code>df -h</code>
<code>du -sh</code>	Folder size	<code>du -sh backend</code>
<code>ls -lh</code>	Human-readable sizes	<code>ls -lh</code>
<code>free -h</code>	RAM usage	<code>free -h</code>
<code>uptime</code>	System uptime/load	<code>uptime</code>
<code>lsblk</code>	Block devices/disks	<code>lsblk</code>

PART 9 — Package Management

Ubuntu/Debian

Command	काम
<code>sudo apt update</code>	Package lists refresh
<code>sudo apt upgrade</code>	Packages upgrade
<code>sudo apt install nginx</code>	Install
<code>sudo apt remove nginx</code>	Remove
<code>sudo apt search nginx</code>	Search
<code>apt show nginx</code>	Package info
<code>sudo apt autoremove</code>	Unused dependencies remove

Amazon Linux/RHEL-family systems

Command	काम
<code>sudo yum update</code>	Update
<code>sudo yum install git</code>	Install
<code>sudo yum remove git</code>	Remove
<code>yum search git</code>	Search
<code>yum list installed</code>	Installed packages
<code>yum info git</code>	Package info

PART 10 — Services

Command	काम
<code>systemctl status nginx</code>	Status
<code>sudo systemctl start nginx</code>	Start
<code>sudo systemctl stop nginx</code>	Stop
<code>sudo systemctl restart nginx</code>	Restart
<code>sudo systemctl enable nginx</code>	Boot पर start
<code>sudo systemctl disable nginx</code>	Boot auto-start बंद
<code>systemctl list-units --type=service</code>	Services की list

🔥 PART 11 — Shell Operators जो इन Commands के साथ सीखने चाहिए

Symbol	काम	Example
	Output → next command input	<code>history grep git</code>
>	Output को file में लिखना/overwrite	<code>ls > files.txt</code>
>>	Output append	<code>ls >> files.txt</code>
&&	Previous success होने पर next command	<code>mkdir app && cd app</code>
;	Commands sequentially execute	<code>pwd; ls</code>
&	Command को background में चलाने के लिए shell behavior	<code>command &</code>
\$()	Command substitution	<code>echo \$(pwd)</code>
*	Wildcard	<code>ls *.js</code>

